

EDS Assignment 2

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◆ **TITLE :** Movie Review Sentiment and Rating Analysis using NumPy and Pandas

◆ **DATASET**

The dataset used in this project is a Movie Review Dataset that contains records of 30 well-known movies from various genres. It captures both the financial and audience perspective of each movie, including details such as the movie title, genre, IMDB rating, number of reviews received, audience sentiment, release year, runtime in minutes, production budget, and box office collection.

-This dataset is useful for studying:

- how movie ratings vary across different genres.
- whether a movie received mostly positive or negative reviews from the audience.
- which genres are most popular and highest rated.
- how production budget impacts the box office collection of a movie.
- trends in movie releases and performance over different years.

-The sentiment column in the dataset is labeled as:

- positive the audience enjoyed the movie and gave a good review
- negative the audience did not enjoy the movie and gave a bad review

The dataset contains 30 rows and 9 columns in total, making it a compact and easy to analyze dataset for performing data analysis operations using NumPy and Pandas.

◆ PROBLEM STATEMENT

The entertainment industry generates large amounts of user review data on platforms like IMDB that can be analysed to understand audience preferences and movie performance trends. This project aims to examine an IMDB Movie Reviews dataset using **NumPy** and **Pandas** in order to study movie characteristics such as ratings, genres, sentiments, and financial performance.

The analysis focuses on finding statistical measures of movie ratings, identifying the count of positive and negative reviews, examining genre-wise and year-wise patterns, and understanding how budget relates to box office collection. It also helps in understanding how ratings and sentiments are distributed across different genres and release years.

◆ CODE (implemented using Google Colab)

```
[3]
✓ 4s from google.colab import files
      uploaded = files.upload() # Upload your IMDB_Dataset.csv file

      import numpy as np
      import pandas as pd

      df = pd.read_csv("IMDB Dataset.csv")

> ... Show hidden output

[4]
✓ 0s # 1.Displaying the first 5 rows to get an overview of the dataset
      print(df.head())

      movie_title  genre  rating  num_reviews  sentiment  release_year \
0  The Dark Knight  Action    9.0         4500    positive        2008
1      Inception    Action    8.8         4200    positive        2010
2  The Notebook    Romance    7.5         2100    positive        2004
3  Titanic          Romance    7.8         3900    positive        1997
4  The Conjuring    Horror    7.5         1800    positive        2013

      runtime_min  budget_million  box_office_million
0             152           185.0           1004.6
1             148           160.0            836.8
2             123            29.0            115.6
3             195           200.0          2187.5
4             112            20.0            319.5

[5]
✓ 0s # 2.Displaying the last 5 rows to check the end of the dataset
      print(df.tail())

      movie_title  genre  rating  num_reviews  sentiment  release_year \
25  A Quiet Place  Horror    7.5         2000    positive        2018
26  Knives Out    Thriller    7.9         2700    positive        2019
27  The Proposal  Romance    6.7         1400    positive        2009
28  Bad Boys      Action    6.9         1500    positive        1995
29  Game Night    Comedy    7.0         1300    positive        2018

      runtime_min  budget_million  box_office_million
25             90            17.0            340.9
26            131            40.0            311.4
27            108            40.0            317.4
28            119            19.0            141.4
29             100            37.0            117.4

[6]
✓ 0s # 3. Display the total number of rows and columns
      print("Total Rows:", df.shape[0])
      print("Total Columns:", df.shape[1])

      Total Rows: 30
      Total Columns: 9

[7]
✓ 0s # 4. Display all column names
      print(df.columns)

      Index(['movie_title', 'genre', 'rating', 'num_reviews', 'sentiment',
```

```
[7] # 4. Display all column names
✓ Os print(df.columns)

Index(['movie_title', 'genre', 'rating', 'num_reviews', 'sentiment',
      'release_year', 'runtime_min', 'budget_million', 'box_office_million'],
      dtype='object')

[8] # 5. Display the data types of each column
✓ Os print(df.dtypes)

movie_title      object
genre            object
rating           float64
num_reviews      int64
sentiment        object
release_year     int64
runtime_min      int64
budget_million   float64
box_office_million float64
dtype: object

[9] # 6. Check for missing values in the dataset
✓ Os print(df.isnull().sum())

... movie_title      0
genre              0
rating            0
num_reviews       0
sentiment         0
release_year      0
runtime_min       0
budget_million    0
box_office_million 0
dtype: int64

[10] # 7. Display basic statistical summary of the dataset
✓ Os print(df.describe())

count      rating  num_reviews  release_year  runtime_min  budget_million \
mean      7.520000   2806.666667   2008.966667   126.200000   65.523333
std       1.283153   1330.784921    10.639819    25.972665   77.664427
min       2.800000    800.000000   1972.000000    88.000000   1.500000
25%       6.925000   1625.000000   2004.750000   108.500000   19.250000
50%       7.700000   2650.000000   2010.000000   121.000000   36.000000
75%       8.475000   3975.000000   2017.750000   141.000000   75.250000
max       9.200000   5000.000000   2019.000000   195.000000   356.000000

count      box_office_million
mean      506.296667
std       603.839243
min       49.000000
25%       240.150000
50%       314.400000
75%       462.150000
max       2797.800000
```

```
[11] # 8. Find the mean rating using NumPy
✓ Os print("Mean Rating:", np.mean(df['rating']))

Mean Rating: 7.52

[12] # 9. Find the median rating using NumPy
✓ Os print("Median Rating:", np.median(df['rating']))

Median Rating: 7.7

[13] # 10. Find the standard deviation of ratings using NumPy
✓ Os print("Standard Deviation:", np.std(df['rating']))

Standard Deviation: 1.2615863030328127

[14] # 11. Find the minimum and maximum rating using NumPy
✓ Os print("Minimum Rating:", np.min(df['rating']))
print("Maximum Rating:", np.max(df['rating']))

Minimum Rating: 2.8
Maximum Rating: 9.2

[15] # 12. Display the count of positive and negative sentiments
✓ Os print(df['sentiment'].value_counts())

... sentiment
positive    25
negative     5
Name: count, dtype: int64

[16] # 13. Display the count of movies in each genre
✓ Os print(df['genre'].value_counts())

genre
Action      6
Romance     5
Horror       5
Drama        5
Comedy       5
Thriller     4
Name: count, dtype: int64

[17] # 14. Display the average rating of each genre
✓ Os print(df.groupby('genre')['rating'].mean())

genre
Action      7.416667
Comedy      6.260000
Drama       8.720000
Horror      7.320000
Romance     7.380000
Thriller    8.175000
Name: rating, dtype: float64

[18]
```

[18]
✓ Os

15. Display the genre with the highest average rating
avg_by_genre = df.groupby('genre')['rating'].mean()
print("Best Genre:", avg_by_genre.idxmax())
print("Average Rating:", avg_by_genre.max())

Best Genre: Drama
Average Rating: 8.72

[19]
✓ Os

16. Display all movies with rating greater than 8
high_rated = df[df['rating'] > 8]
print(high_rated[['movie_title', 'genre', 'rating']])

	movie_title	genre	rating
0	The Dark Knight	Action	9.0
1	Inception	Action	8.8
6	Avengers Endgame	Action	8.4
7	Joker	Drama	8.5
12	Gone Girl	Thriller	8.1
13	Se7en	Thriller	8.6
14	Parasite	Drama	8.6
15	Whiplash	Drama	8.5
21	Shutter Island	Thriller	8.1
22	The Godfather	Drama	9.2
23	Forrest Gump	Drama	8.8

[20]
✓ Os

17. Display all movies with positive sentiment only
positive_movies = df[df['sentiment'] == 'positive']
print(positive_movies[['movie_title', 'genre', 'rating']])

...	movie_title	genre	rating
0	The Dark Knight	Action	9.0
1	Inception	Action	8.8
2	The Notebook	Romance	7.5
3	Titanic	Romance	7.8
4	The Conjuring	Horror	7.5
6	Avengers Endgame	Action	8.4
7	Joker	Drama	8.5
8	La La Land	Romance	8.0
9	Get Out	Horror	7.7
10	The Hangover	Comedy	7.7
11	Superbad	Comedy	7.6
12	Gone Girl	Thriller	8.1
13	Se7en	Thriller	8.6
14	Parasite	Drama	8.6
15	Whiplash	Drama	8.5
20	The Ring	Horror	7.1
21	Shutter Island	Thriller	8.1
22	The Godfather	Drama	9.2
23	Forrest Gump	Drama	8.8
24	Crazy Rich Asians	Romance	6.9
25	A Quiet Place	Horror	7.5
26	Knives Out	Thriller	7.9
27	The Proposal	Romance	6.7
28	Bad Boys	Action	6.9
29	Game Night	Comedy	7.0

[21]
✓ Os

18. Display movies released after the year 2015

[21]
✓ Os

18. Display movies released after the year 2015
recent_movies = df[df['release_year'] > 2015]
print(recent_movies[['movie_title', 'release_year', 'rating']])
print("Total Count:", len(recent_movies))

	movie_title	release_year	rating
6	Avengers Endgame	2019	8.4
7	Joker	2019	8.5
8	La La Land	2016	8.0
9	Get Out	2017	7.7
14	Parasite	2019	8.6
17	Cats	2019	2.8
18	The Mummy	2017	5.4
24	Crazy Rich Asians	2018	6.9
25	A Quiet Place	2018	7.5
26	Knives Out	2019	7.9
29	Game Night	2018	7.0

Total Count: 11

[22]
✓ Os

19. Display the movie with the highest rating
print(df[df['rating'] == df['rating'].max()][['movie_title', 'rating']])

	movie_title	rating
22	The Godfather	9.2

[23]
✓ Os

20. Display the movie with the lowest rating
print(df[df['rating'] == df['rating'].min()][['movie_title', 'rating']])

...	movie_title	rating
17	Cats	2.8

[24]
✓ Os

21. Display movies with positive sentiment and rating greater than 7
positive_high = df[(df['sentiment'] == 'positive') & (df['rating'] > 7)]
print(positive_high[['movie_title', 'genre', 'rating', 'sentiment']])
print("Total Count:", len(positive_high))

...	movie_title	genre	rating	sentiment
0	The Dark Knight	Action	9.0	positive
1	Inception	Action	8.8	positive
2	The Notebook	Romance	7.5	positive
3	Titanic	Romance	7.8	positive
4	The Conjuring	Horror	7.5	positive
6	Avengers Endgame	Action	8.4	positive
7	Joker	Drama	8.5	positive
8	La La Land	Romance	8.0	positive
9	Get Out	Horror	7.7	positive
10	The Hangover	Comedy	7.7	positive
11	Superbad	Comedy	7.6	positive
12	Gone Girl	Thriller	8.1	positive
13	Se7en	Thriller	8.6	positive
14	Parasite	Drama	8.6	positive
15	Whiplash	Drama	8.5	positive
20	The Ring	Horror	7.1	positive
21	Shutter Island	Thriller	8.1	positive
22	The Godfather	Drama	9.2	positive
23	Forrest Gump	Drama	8.8	positive
25	A Quiet Place	Horror	7.5	positive

```
[25]
✓ Os # 22. Display the average budget and average box office collection
print("Average Budget (million):", df['budget_million'].mean().round(2))
print("Average Box Office (million):", df['box_office_million'].mean().round(2))

Average Budget (million): 65.52
Average Box Office (million): 506.3

[26]
✓ Os # 23. Display total box office collection of all movies using NumPy
print("Total Box Office Collection (million):", np.sum(df['box_office_million']))

... Total Box Office Collection (million): 15188.9

[27]
✓ Os # 24. Display the number of movies released each year
print(df['release_year'].value_counts().sort_index())

release_year
1972      1
1994      1
1995      2
1997      1
2000      1
2002      1
2004      1
2007      2
2008      1
2009      2
2010      3
2013      1
2014      2
2016      1
2017      2
2018      3
2019      5
Name: count, dtype: int64

[28]
✓ Os # 25. Sort the dataset by rating in descending order and display top 10
sorted_df = df.sort_values(by='rating', ascending=False)
print(sorted_df[['movie_title', 'genre', 'rating']].head(10))

movie_title  genre  rating
22  The Godfather  Drama    9.2
0   The Dark Knight  Action    9.0
23  Forrest Gump   Drama    8.8
1   Inception     Action    8.8
13  Se7en         Thriller  8.6
14  Parasite      Drama    8.6
15  Whiplash      Drama    8.5
7   Joker         Drama    8.5
6  Avengers Endgame  Action    8.4
21  Shutter Island  Thriller  8.1

[29]
✓ Os # 26. Display the average runtime of all movies using NumPy
print("Average Runtime (minutes):", np.mean(df['runtime_min']))

Average Runtime (minutes): 126.2
```

```
[30]
✓ Os # 27. Display the genre with the most number of movies
print("Genre with Most Movies:", df['genre'].value_counts().idxmax())
print("Count:", df['genre'].value_counts().max())

Genre with Most Movies: Action
Count: 6

[31]
✓ Os # 28. Display all movies where budget is greater than 100 million
big_budget = df[df['budget_million'] > 100]
print(big_budget[['movie_title', 'genre', 'budget_million']])
print("Total Count:", len(big_budget))

movie_title  genre  budget_million
0   The Dark Knight  Action    185.0
1   Inception       Action    160.0
3   Titanic         Romance    200.0
6  Avengers Endgame  Action    356.0
16  Transformers    Action    150.0
18  The Mummy       Action    125.0
Total Count: 6

[32]
✓ Os # 29. Display the count of positive and negative sentiments genre wise
print(df.groupby(['genre', 'sentiment'])['movie_title'].count())

... genre  sentiment
Action  negative    2
        positive    4
Comedy   negative    2
        positive    3
Drama    positive    5
Horror   negative    1
        positive    4
Romance  positive    5
Thriller positive    4
Name: movie_title, dtype: int64
```

[33]
✓ Os

```
# 30. Display movies where box office collection is more than the budget
profit_movies = df[df['box_office_million'] > df['budget_million']]
print(profit_movies[['movie_title', 'budget_million', 'box_office_million']])
print("Total Profitable Movies:", len(profit_movies))
```

▼

	movie_title	budget_million	box_office_million
0	The Dark Knight	185.0	1004.6
1	Inception	160.0	836.8
2	The Notebook	29.0	115.6
3	Titanic	200.0	2187.5
4	The Conjuring	20.0	319.5
5	Insidious	1.5	99.0
6	Avengers Endgame	356.0	2797.8
7	Joker	55.0	1079.0
8	La La Land	30.0	446.1
9	Get Out	4.5	255.4
10	The Hangover	35.0	467.5
11	Superbad	20.0	170.8
12	Gone Girl	61.0	369.3
13	Se7en	33.0	327.3
14	Parasite	11.4	258.8
15	Whiplash	3.3	49.0
16	Transformers	150.0	709.7
18	The Mummy	125.0	409.2
19	Scary Movie	19.0	278.0
20	The Ring	48.0	249.3
21	Shutter Island	80.0	294.8
22	The Godfather	6.0	245.1
23	Forrest Gump	55.0	678.2
24	Crazy Rich Asians	30.0	238.5
25	A Quiet Place	17.0	340.9
26	Knives Out	40.0	311.4
27	The Proposal	40.0	317.4
28	Bad Boys	19.0	141.4
29	Game Night	37.0	117.4

Total Profitable Movies: 29